

SALES FORECASTING & DEMAND INTELLIGENCE

Executive Business Report

Prepared for: Head of Supply Chain & Chief Financial Officer | Data Period: Jan 2015 – Dec 2018 | Prepared by: Divesh Matkar, AI/Data Science

Executive Summary

Over the past four years, the business generated \$2.26 million in sales across 9,800 orders. Technology, Furniture, and Office Supplies each contribute a meaningful and fairly even share of revenue, with the West region leading and the South region lagging. Looking ahead, our forecasting model expects total sales over the next three months to land around \$152,000, though this comes with real uncertainty that the business should plan around rather than treat as a fixed number. We also found clear, explainable spikes and dips in weekly sales tied to the holiday shopping season, and we grouped products into four demand groups so that stocking decisions can be made by group rather than product-by-product guesswork. Acting on these findings can reduce both stockouts during peak periods and excess inventory in slow-moving categories.

Key Findings from EDA & Forecasting

- Category mix is balanced.** Technology is the top-earning category at \$827K (37% of total sales), followed by Furniture (\$729K) and Office Supplies (\$705K) — the three categories are closer in size than a typical retailer might expect.
- Regional performance is uneven.** West (\$710K) and East (\$670K) generate nearly 60% of all sales, while South trails at \$389K, suggesting an opportunity to investigate regional demand or distribution gaps.
- Consumer segment dominates.** Consumer accounts for \$1.15M (51%) of sales versus \$688K for Corporate and \$425K for Home Office — Consumer-focused strategies will move the most revenue.
- Two products carry outsized weight.** Phones (\$328K) and Chairs (\$323K) are the single highest-earning sub-categories and warrant dedicated forecasting and stock priority.
- Sales follow a predictable seasonal rhythm.** A statistical stationarity check (ADF test) confirmed that monthly sales follow a stable, repeatable pattern rather than a random walk, which means historical patterns are a reliable basis for forecasting.

3-Month Sales Forecast

Using a seasonal forecasting model (SARIMA) trained on four years of monthly sales, we project the following sales for the three months immediately after the dataset ends. The “likely range” reflects our 95% confidence band — in plain terms, we expect the actual number to fall inside that range about 19 times out of 20.

Month	Expected Sales	Likely Range (95% confidence)
Month 1	\$45,900	\$18,900 – \$73,000
Month 2	\$37,100	\$10,000 – \$64,200
Month 3	\$69,000	\$42,000 – \$96,100

In plain language: we expect a modest start to the quarter, a slight dip in month two, and a rebound in month three that likely reflects the same seasonal upswing seen in prior years. Total expected sales for the quarter are approximately \$152,000, but the wide ranges above mean the business should build plans that work even if actual sales land near the low end.

Top 3 Sales Anomalies Detected

- 1. Pre-Black Friday surge (mid-November 2018):** A weekly sales spike of roughly \$30,600 — about 3x the typical week — was detected in the run-up to Black Friday, most likely reflecting early holiday promotions and pre-season stocking by customers.
- 2. Cyber Monday / holiday peak (early December 2018):** The single highest sales week in the four-year dataset (~\$36,000) occurred at the start of December, consistent with Cyber Monday and the peak of holiday gift purchasing.

- **3. Post-holiday slump (early January 2015):** The lowest sales week in the entire dataset (under \$1,000) occurred in early January, right after the holiday season — a common post-holiday demand lull combined with lighter order volume at the very start of the data collection period.

Both detection methods used (Isolation Forest and rolling Z-score) independently flagged these events, which increases our confidence that they reflect real, explainable business patterns rather than data errors.

Product Demand Segmentation & Stocking Strategy

Products were grouped using clustering based on total sales, average order value, and order frequency. Four distinct demand groups emerged:

Segment	Products	Recommended Stocking Strategy
Core Revenue Drivers	Accessories, Chairs, Phones, Storage, Tables	High, always-in-stock inventory with frequent replenishment; these carry the most sales volume and the highest stockout risk.
Premium / Big-Ticket	Copiers, Machines	Low stock, high value per unit (avg. ~\$1,930/order); favor made-to-order or just-in-time supply to limit capital tied up in inventory.
High-Frequency Staples	Binders, Furnishings, Paper	Low unit value but very frequent orders; automate reordering with bulk purchasing to keep unit costs down.
Low-Priority / Niche	Appliances, Art, Bookcases, Envelopes, Fasteners, Labels, Supplies	Minimal stock, order-on-demand; lowest revenue contribution, and a candidate list for SKU rationalization.

Business Recommendations

- **1. Protect stock levels for Core Revenue Drivers.** Prioritize replenishment budget for Accessories, Chairs, Phones, Storage, and Tables ahead of the next quarter. These five sub-categories carry the highest sales volume and order frequency in the data, and the 3-month forecast shows continued demand.
- **2. Build a holiday surge plan starting in October.** Because two of the year's three biggest anomalies were holiday-driven spikes (mid-November and early December), build inventory and staffing buffers 3–4 weeks ahead of the holiday season rather than reacting to it in real time.
- **3. Reduce inventory investment in Low-Priority / Niche products.** Appliances, Art, Bookcases, Envelopes, Fasteners, Labels, and Supplies together contribute the least total revenue relative to the shelf space and capital they occupy. A SKU review of this group — consolidating suppliers or shifting to drop-ship — can free up working capital for the higher-performing segments above.

Risk & Limitation to Be Aware Of

The forecasting model's accuracy, measured on the most recent three known months, showed an average error of about 22% (MAPE). This is a reasonable starting point for planning but not precise enough for tight, single-number budgeting — the ranges in the forecast table should be treated as the real answer, not the midpoint. The model also does not account for external factors such as planned promotions, competitor pricing, or broader economic shifts, since it is built purely from historical sales patterns. We recommend re-validating the forecast monthly as new sales data arrives and pairing it with input from category managers before finalizing purchase orders.